

Olam Agri: choices that remain useful to 2051

Commit first to customer proof, route cash and adaptable capacity.

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Independent strategic research; company disclosures, external projections and illustrative models are distinguished.

Approve evidence before irreversible expansion

Authorize a scoped 90-day proof programme; defer an unconditional asset cheque.

Next commitment	Required evidence	Owner
Data room	Eligible sales, cost, capacity, seasonal AR/INV/AP	Finance + operations
Pilot	Pre-registered counterfactual and downside limits	Country + marketing
Investment review	Tax, route, resilience, funded cash and positive NPV	Treasury + IC

Finance owns the data room, costed pilot envelope and cash limits. No unsupported funding amount is requested. Nigeria is the initial commercial laboratory; select one sourcing corridor and one transfer market after feasibility checks. Only customer-level contribution, funding headroom and replicable learning can release a later commitment.

The old Nigeria screen becomes a global decision system

The negative legacy case remains visible; additional detail cannot manufacture approval.

Starting point	Replacement
44-slide Nigeria-first narrative	37-slide global executive sequence + technical appendix
Sales-day cash proxy	AR + cost-based inventory – eligible purchases AP
Three-year uplift screen	25 annual cash periods with replacement, tax and ramp
Independent chosen draws	Common-shock stress; no success-probability claim
Opaque review-count claim	Versioned evidence, calculation IDs and actual checks

Volume growth did not translate into stronger EBIT

FY2025 Olam Agri: revenue +15.4%, sales volume +19.1%, EBIT –8.0%.

USD reporting perimeter	2024	2025
Revenue, USD million	24,848	28,666
Sales volume, thousand tonnes	45,130	53,740
EBIT, USD million	765.2	703.7
Derived EBIT margin	3.08%	2.45%

Customer orders, physical supply and cash must agree

Population creates context; sold tonnes require accessible demand, capture and delivery.

Layer	Evidence required	Capital implication
Customer	Usable yield, cash ticket, counterfactual repeat	Test before scaling
Route	HS access, landed cost, lead time, counterparty	Contract before stocking
Asset	Utilisation, uptime, conversion and resilience	Option capacity only after proof

Demand: households, bakers, industrial buyers and feed customers buy different services.

Supply: origin, quality, yield, logistics and usable capacity bound delivered volume.

Cash: payment, stocks, currency, replacement and tax can overturn a positive gross spread.

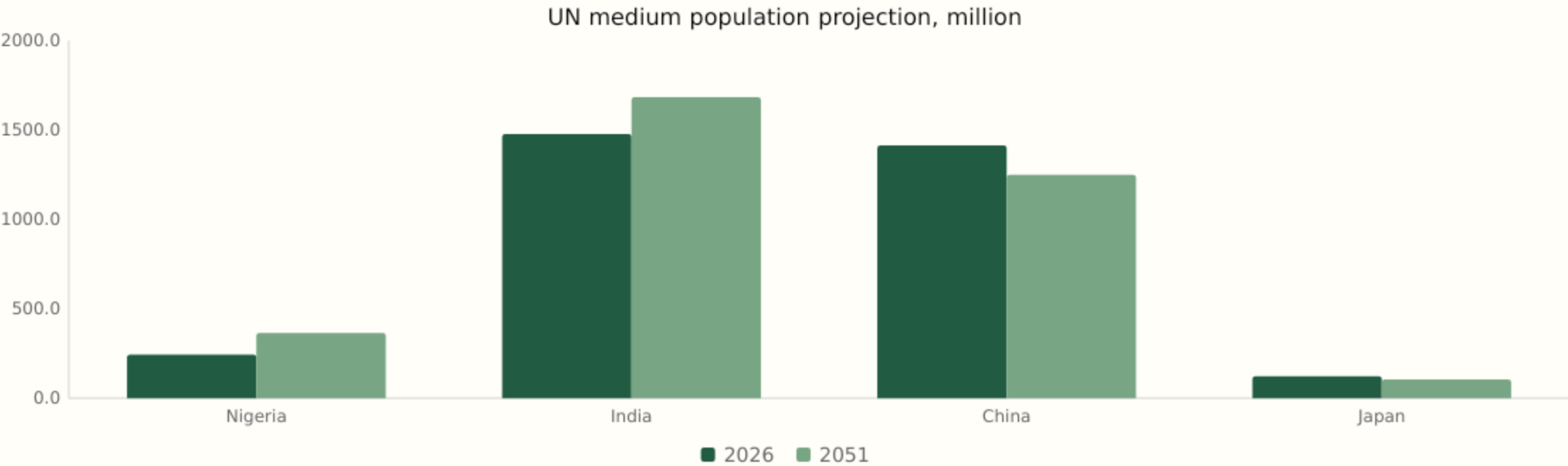
A 25-year horizon strengthens options, not precision

Separate the official 2035 outlook from our conditional 2036–2051 extension.

Period	What is evidenced	Decision rule
2026–2030	Present operations and near-term customer tests	Repair measured bottlenecks
2030–2035	OECD–FAO outlook; UN population paths	Choose modular capacity after contracted cash proof
2036–2051	UN demography + chosen bounded operating paths	Refresh annually; commit against triggers
Every stage	No guaranteed terminal franchise value	Fund closeout and replacement explicitly

Population pathways diverge; category demand remains unmeasured

UN medium population projections do not imply equal per-person purchases.



SOURCE UN_WPP · WB_income_ppp · WB_urban

Sell customer economics before asserting a nutrition premium

Health and convenience are hypotheses until customers sustain paid repeat.

Buyer	Testable value	Adoption constraint
Household	Cost per meal, cash ticket and preparation	Income, trust and availability
Baker	Saleable output, rejects and energy per bag	Working cash and consistency
Industrial buyer	Specification, traceability and delivery	Contract economics and substitution
Feed customer	Verified performance per animal output	Credit, formulation and biological variance

Country averages cannot price asset-level climate exposure

Map crop, basin, route and facility hazards before estimating damage or adaptation returns.

Exposure	Response to test
Common harvest + port stress	Alternate origin with genuinely different exposure
Water constraints	Basin-specific supplier practices and verified savings
Flood/heat asset downtime	Maintenance, redundancy and insurance economics

IPCC supports physical risk mechanisms; WMO global indicators do not yield local output losses.
Water-stress observations help diligence, but suppliers may draw from different basins.
Require crop calendars, asset coordinates, downscaled hazards, adaptation cost and service losses.

A large import bill is neither accessible demand nor Olam revenue

Routes require product-level permissions, quality, landed spreads and counterparties.

Nigeria wheat preview: 2024 CIF value about USD1.0475 bn; missing net weight prevents a credible tonnes/unit-price calculation.

Comtrade aggregation and reporting flags remain visible. A quantity field of zero with an unspecified unit is not zero imports.

Sanctions, export restrictions, duties, EUDR and documentation are gates, not scalar scores.

Each value chain needs its own offer and cash model

June-2026 disclosed scope includes food, feed and non-food agricultural products.

Value chain	Initial commercial mechanism	Key test
Grains, rice, wheat milling/pasta	Quality, usable yield and reliable delivery	Landed spread + repeat
Edible oils	Local cash ticket and verified refining economics	Crude/refined HS + conversion
Feed/protein	Performance-based technical service	Buyer margin + biology
Cotton, wood, rubber	Specification, traceability, industrial contracts	Eligibility + funding
Sugar/biofuel	Conditional scope in June factsheet	Current perimeter + policy + process

All 195 countries have a disposition; none has an internal-data valuation

162 countries have all 11 macro fields available at some date; this is not investment readiness.

Coverage test	Result
UN members + observer states	$193 + 2 = 195$
Country $\times$ scenario $\times$ milestone rows	$195 \times 6 \times 7 = 8,190$
Priority diligence dossiers	10
Investment-grade country cases	0
Remaining countries	Watch, existing-operation review, defer or no standalone asset proposal

SOURCE UN_M49 - UN_WPP

Choose ten diligence cases, with distinct operating reasons

These are evidence priorities, not a numerical attractiveness ranking.

Markets	Product / mode	Next falsifier
Nigeria, Ghana, Senegal	Semolina/oils; flour/pasta/feed service pilot	Customer and route contribution
Cameroon, Mozambique	Flour/rice/oils distribution pilot	Credit, resilience and cash ticket
South Africa	B2B grains/oils contract trial	Quality-adjusted landed spread
India, Vietnam, Brazil	Rice or grains/oilseeds origination	Supplier, basin and delivered cash
Saudi Arabia	Bankable commercial offtake diligence	Buyer commitment; no ownership-derived order

Managed transition is a planning convention, not the most likely future

Customer orders, capacity, stocks and taxes drive a finite real-USD example.

Illustrative owned-processing input	Managed case
Orders / physical capacity	100,000 / 150,000 tonnes per year
Price / variable cost	\$400 / \$340 per tonne
Capex / fixed cash cost	\$20 m / \$2 m per year
Ramp / growth	50%, 80%, 100%; 1.5%, then 0.75%
Hurdle / tax convention	10% real USD / 25% with loss carryforward

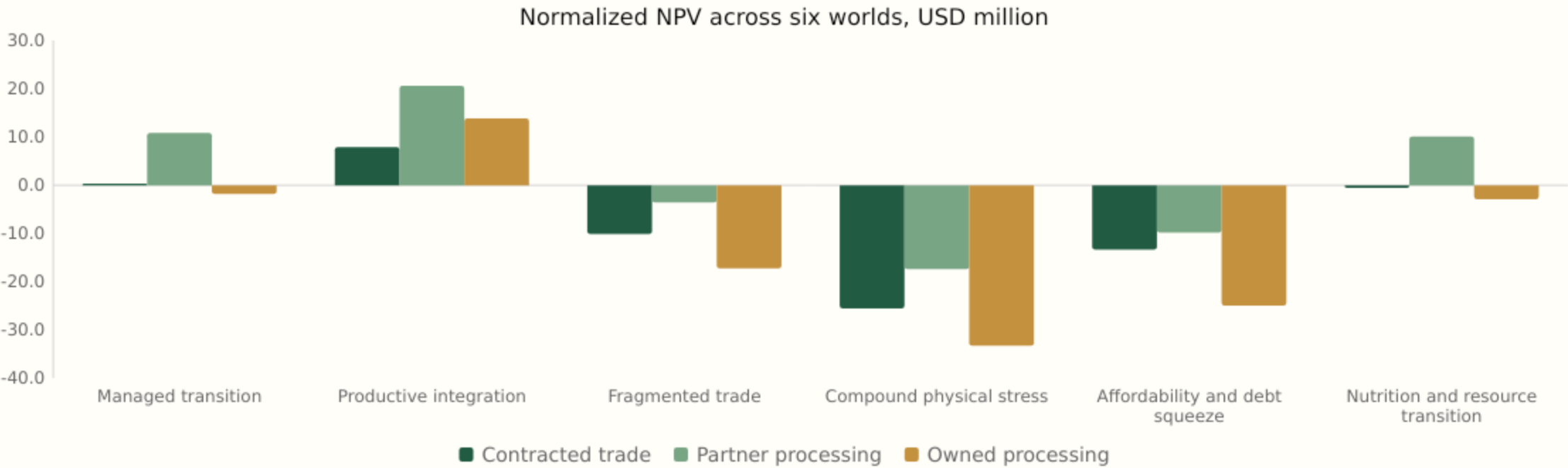
Six coherent worlds alter both customers and the cash cycle

No calibrated probability is assigned to any complete 25-year path.

World	Order factor	Unit cost factor	Cash-days / hurdle
Managed transition	1.0	1.0	+0 / 10%
Productive integration	1.12	0.985	-5 / 9%
Fragmented trade	0.9	1.07	+15 / 12%
Compound physical stress	0.82	1.15	+25 / 12%
Affordability and debt squeeze	0.82	1.04	+20 / 14%
Nutrition and resource transition	1.03	1.02	+5 / 10%

No fixed project survives every tested world without adaptation

The financial model deliberately continues fixed commitments to expose failure.



An investment boundary is more useful than a point forecast

Test order thresholds and liquidity limits before expanding physical commitments.

Managed template	Operating break-even t	Zero-NPV initial orders t
Contracted trade	31,250.00	89,195.83
Partner processing	25,000.00	62,437.79
Owned processing	33,333.33	104,862.13

India/Vietnam-to-Africa rice warrants a contracted-lot test

Company financing disclosure supports a corridor, not its incremental returns.

Annual report describes USD100 m seven-year FMO financing associated with rice flows from India, Thailand and Vietnam to Africa.

Choose a destination only after offtake, freight, quality, HS rules and payment are documented.

Compare delivered cash through a small contracted lot; map origin and destination common exposures.

Saudi offtake and Brazil origination require different proof

SALIC control offers strategic alignment; it does not create guaranteed sales.

Case	Why examine it	What prevents approval
Saudi Arabia	Disclosed location and strategic shareholder	No verified incremental bankable order
Brazil	Disclosed grain/oilseed origin/export capability	No verified buyer spread, supplier/basin or port risk
Ownership	81.81% SALIC; 18.19% Olam Group after CFG	About page lower 81.11% conflict recorded

The inherited Nigeria capital screen stays on hold

NGN0.47bn annual uplift does not fund NGN4.2bn initial capital at the chosen hurdle.

Inherited screening assumption / result	NGN billion
Sales base / contribution margin	100 / 8%
Net annual uplift after chosen leakage	0.470
One-time 7 sales-day proxy release	1.918
Capital envelope	4.200
Three-year NPV at 28%	−2.065
Zero-NPV annual uplift	1.993

Customer value must turn into paid, profitable repeat

A lower cost per usable output can justify price only if cash and trust permit adoption.

Illustrative flour offer	A	B
Bag price + preparation cost	\$30 + \$5	\$32 + \$4
Usable output, kg	45	48
Cost per usable kg	\$0.778	\$0.750
B advantage	—	3.57% lower output cost

Small packs solve a cash ticket, not necessarily unit value

Test price, usable yield, availability and channel economics as separate mechanisms.

Intervention	Customer test	Olam cash test
Pack size	Cash outlay + cost per meal	Net price, packing and route cost
Baker technical service	Saleable output, rejects and energy	Contribution after service
Route availability	Paid repeat and substitution	Stock turns + credit losses
Price recovery	Pass-through timing and demand response	Retained margin, not gross price rise

Count only delivered saleable units

The lowest physical or commercial constraint determines revenue.

Illustrative demand chain	Tonnes
1 m people ×100 kg/person/year	100,000 category demand
50% accessible ×20% capture	10,000 buyer orders
Saleable capacity	8,000
Route delivery limit	7,000
Delivered sales	7,000, not 10,000

Finance receivables, inventory and payables on their own bases

Inventory and purchases use cost; receivables use credit sales.

Illustrative release, USD m	Calculation	Result
Receivables	$40 \times 5 / 365$	0.548
Inventory	$34 \times 7 / 365$	0.652
Payables	DPO unchanged	0.000
Total once-only transition cash	$0.548 + 0.652$	1.200

Positive EBITDA can coexist with negative investment NPV

Managed owned-processing template: NPV –USD1.80m; peak annual funding USD22.56m.

Full model convention	Effect
Initial capex	60% now +40% in first operating year
Maintenance / replacement	2% annually; 70% in years 11 and 21
Tax / depreciation	25% simplified carryforward; capital cohorts over 10 years
Working cash	80% credit-sales AR + cost inventory –80% eligible-purchases AP
End of 2051	90% NWC closeout; no franchise value

Common shocks can consume more cash than the asset cheque

Chosen stress distribution illustrates failure; it does not estimate the probability of loss.

Owned-template stress, USD m	10,000 draws
10 th /50 th /90 th percentile NPV	-19.27 / -1.65 / 21.48
Loss VaR95	24.54
Loss CVaR95	29.25
Managed zero-NPV order boundary	104,862 tonnes of initial annual orders
Response	Reprice, resize, delay or stop; never assume automatic recovery

Diversification must reduce common exposure, not add flags

Allocate only among legally feasible, funded and physically deliverable choices.

Benchmark	Failure to test
Population / import-size ranking	Capture, spread, seasonal funding
Expected NPV only	Joint tail loss and cash breach
Country count diversification	Shared port, crop season or lender

Group projects by crop/season, basin, port, shipping lane, buyer, FX and lender before optimizing.
Use budget, peak cash, capacity, water/carbon and counterparty constraints with explicit shadow costs.
No optimal all-country allocation is claimed without country economics and dependence data.

Reassess choices as customer and route evidence arrives

The proposed model combines established methods into an executable decision workflow.

Step	Operation
1. Feasible set	Product access, ethics, rights, legal and cash constraints
2. Operating model	Demand → delivery → cash in six conditional worlds
3. Robust action	Minimize worst regret subject to capital and liquidity
4. Learning choice	Study only if net decision value exceeds cost/delay
5. Policy trigger	Invest, wait, reprice, resize, stop or exit

Robust regret can choose a partner over the largest plant

A transparent teaching case changes the decision under the same USD20m maximum envelope.

Chosen payoff, USD m	Managed	Integration	Physical	Max regret
Wait	0.0	0.0	0.0	14.0
Large owned plant	9.0	14.0	-8.0	8.0
Partner processing	6.0	10.0	-1.0	4.0
Capped contracted trade	4.0	7.0	0.0	7.0

The executed supply regression is insufficient for investment forecasting

A small error gain does not turn aggregate food production into product demand.

Validation, annual log growth	Result
Train / held-out records	2,856 /680
Holdout period	2019-2022
Model MAE /country-mean MAE	0.04830 /0.04979
In-sample R^2	0.00618
Selection	Reject for investment forecasting; retain benchmarks

SOURCE WB_food_index · WB_income_ppp · WB_urban · WB_cereal_yield

Pre-register contribution and cash, then test incremental marketing

Use randomised route clusters when feasible; otherwise defend the counterfactual.

Pilot requirement	Concrete specification
Eligible outcome	Paid sell-through × incremental contribution, net of service/trade cost
Counterfactual	Holdout routes with baseline and spillover checks
Sizing	Baseline variance, detectable effect, clustering and attrition
Decision	Confidence interval + contribution + funding + sustained repeat
No claim today	No measured Olam treatment effect or causal ROMI

Dates organize review; evidence releases commitments

Milestones through 2051 are review windows, not promises of expansion.

Window	Capability	Release condition
2026-2030	Customer data, route P&L and seasonal cash	Replicated contribution and legal clearance
2030-2035	Supplier diversity and modular processing	Contracted demand, financed cash and resilient delivery
2035-2045	Reconfigure origins, assets and offers	Updated hazards, customer preferences and opportunity cost
2045-2051	Renew, repurpose or close capacity	Replacement economics and funded orderly exit

G0–G7 gives every commitment an owner and a stop rule

Legal and liquidity failures override an attractive demand story.

Gate	Owner / release evidence
G0 perimeter + G1 access	Strategy/Legal: right entity, HS, sanctions, rights and licenses
G2 customer + G3 contribution	Marketing/Finance: paid repeat and costed counterfactual
G4 capacity + G5 resilience	Operations: throughput, yield, route and common-shock response
G6 cash + G7 replication	Treasury/IC: seasonal cash, stress funding and transfer proof
Monitor / stop	Monthly operational signals; immediate breach escalation

I can connect customer experiments to a traceable cash decision

My offer is an operating measurement system with Finance-controlled economics.

Deliverable I can own	Accountability
Data dictionary + experiment design	Reconcile eligible records with Finance
Country/customer evidence system	Version sources and falsifiers
Commercial proof-to-cash reporting	Separate causal sales, service cost and stock funding

MBA at IIM Shillong and chemical engineering combine commercial framing with conversion, constraint and process reasoning.
Portfolio evidence records a ₹20 m budget and CPM reduction from ₹35.8 to ₹15.5: 56.7% lower CPM. It does not establish Olam sales, profit or category-management outcomes.
Meghalaya work was a simulation informed by government records; no realised savings are asserted.

Ninety days can earn a better decision, not prove a 25-year forecast

Cost and size the pilot only after inspecting baseline variance and available routes.

Days	Work product	Acceptance
1-15	Data room, entity and customer/route baseline	Finance reconciliation and scope signed
16-30	Pre-registration, power and controlled rollout plan	Economically detectable effect and bounded cash cost
31-60	Execute, audit treatment and delivery	Track attrition, spillover, service and stock
61-75	Cash proof and one transferability test	Repeat after support; contribution interval
76-90	Independent review: scale, resize, wait or stop	Decision memo with falsifiers and funded exposure

Approve the evidence programme and challenge its release criteria

No country asset approval is supported by the present public-data package.

Request customer orders, product/route cost, plant constraints, monthly inventory, AR/AP, tax and currency/hedge records.

Ask country leads to validate ten diligence cases; all other countries retain their explicit atlas disposition.

Set the costed pilot cap, cash-breach limit and minimum economically material effect with Finance.

Return in 90 days with measured contribution, repeat, funding and a specific expand/wait/stop proposal.

Thank you

Questions about the assumptions are the next evidence commitment.

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Complete country atlas, editable workbook, source book and reproducible code accompany this presentation.

The technical appendix begins on the next slide.

Technical appendix / how to read this package

Source IDs, assumption IDs and calculation IDs travel with every decision.

Reported facts: company perimeter and dated primary statistics. External projections: UN medium population and OECD-FAO through 2035.

Derived values: arithmetic from sourced inputs. Chosen scenarios/templates: assumptions, not forecasts. Unknown values remain blank.

C-E01–C-E22 teach arithmetic. C-DCF follows full cash; C-OLS records executed validation; C-STRESS and C-REGRET record chosen decision examples.

01 / Reported margin

2.45% derived EBIT margin

Reported/derived evidence

$$703.7 / 28,666 \times 100 = 2.4548\%$$

Using rounded revenue of USD28.7 bn gives 2.4519%; keep the same entity, period and currency.

Revenue can grow while margin falls.

02 / Demand to delivered sales

7,000 delivered tonnes

Illustrative example—not Olam data

$1 \text{ m people} \times 0.1 \text{ t} \times 50\% \text{ access} \times 20\% \text{ capture} = 10,000 \text{ t orders}$; $\min(10,000, 8,000, 7,000) = 7,000 \text{ t}$.

Demand, access and capture are chosen inputs. Unmet orders cannot count as revenue.

If route capacity reaches 9,000 t, plant capacity still caps delivery at 8,000 t.

03 / Volume, revenue and EBITDA

USD4m EBITDA

Illustrative example—not Olam data

$100,000 \text{ t} \times \text{USD}400/\text{t} = \text{USD}40 \text{ m}$ revenue; costs = USD34 m; contribution = USD6 m; less USD2 m fixed cost = USD4 m EBITDA.

EBITDA omits cash tax, capital spending and working cash. These are chosen saleable-volume and net-price inputs.

A USD10/t price change at unchanged volume changes EBITDA by USD1 m.

04 / Yield and by-products

111,111 tonnes of input

Illustrative example—not Olam data

100,000 t output / 90% yield = 111,111 t input; 6% by-product = 6,667 t; 4% loss = 4,444 t.

By-products may earn revenue. Their quality, price and conversion cost need evidence.

At 95% main yield, input is 105,263 t. Reconcile the remaining mass separately.

05 / Customer cost per usable output

B costs 3.57% less per usable kg

Illustrative example—not Olam data

Offer A: $(\text{USD}30 + \text{USD}5) / 45 \text{ kg} = \text{USD}0.7778/\text{kg}$. Offer B: $(\text{USD}32 + \text{USD}4) / 48 \text{ kg} = \text{USD}0.7500/\text{kg}$.

Chosen flour-like inputs. Paid adoption also depends on cash, trust and availability.

If B produces only 45 kg, cost rises to USD0.80/kg and reverses the advantage.

06 / Pack cash and cost per meal

Small cash ticket ~~₦~~600; large pack 14.29% cheaper per meal

Illustrative example—not Olam data

Small: (~~₦~~600 + ~~₦~~100 prep) / 5 meals = ~~₦~~140/meal. Large: (~~₦~~2,000 + ~~₦~~400 prep) / 20 meals = ~~₦~~120/meal.

Chosen Nigerian cash units and serving size. The large pack needs more upfront cash.

Large purchase cash is 3.33 times the small pack, despite lower cost per meal.

07 / Percent versus percentage points

2 percentage points; 25% relative increase

Illustrative example—not Olam data

$10\% - 8\% = 2$ percentage points. Relative increase = $(10 - 8) / 8 = 25\%$.

A margin change is not the same measure as revenue growth.

On USD100 revenue, profit rises from USD8 to USD10.

08 / Profit to free cash

USD1.25m free cash flow

Illustrative example—not Olam data

EBITDA 4 – D&A 1 = EBIT 3; tax = 0.75. FCFF = $3 - 0.75 + 1 - 0.6 - 0.4 - 1 = 1.25$ (USDm).

Assume no prior tax losses, 25% tax, USD0.6 m maintenance and USD0.4 m growth capex.

If NWC does not increase, this period's cash rises by USD1 m.

09 / Working-capital release

USD1.20m once-only cash release

Illustrative example—not Olam data

AR: $\text{USD}40 \text{ m} \times 5 / 365 = \text{USD}0.547945 \text{ m}$. Inventory: $\text{USD}34 \text{ m} \times 7 / 365 = \text{USD}0.652055 \text{ m}$. Total = $\text{USD}1.20 \text{ m}$.

All sales are credit in this example. The transition release is not recurring annual profit.

One further DSO day releases $\text{USD}0.109589 \text{ m}$ at this sales base.

10 / Inherited Nigeria screen

NPV –NGN2.0645bn; zero-NPV uplift NGN1.9928bn/year

Illustrative example—not Olam data

$100 \times (2\% + 2\%) \times 8\% + 0.7 + 0.6 - 0.4 - 0.45 - 0.30 = 0.47$; $100 \times 7 / 365 = 1.9178$ (NGNbn).

NGN4.2 bn initial capital; 28% hurdle; 50/80/100% benefit ramp. Tax, replacement and actual cash bases are absent.

Repair procurement FX overlap, AR/inventory/AP, capex, tax and currency before a production decision.

11 / FX and commodity interaction

21% combined input-cost increase

Illustrative example—not Olam data

USD100 × ₦1,000/USD = ₦100,000. With both factors +10%: USD110 × ₦1,100 = ₦121,000, or +21%.

Both shocks hit the same procurement base. Hedge effects need documented contracts.

Chosen 70% cost pass-through recovers 14.7% initially; weaker customer volume may reduce retained margin.

12 / Finite NPV and commissioning

NPV –USD0.0263m

Illustrative example—not Olam data

$$\text{NPV} = -5 + 2/1.10 + 2/1.10^2 + 2/1.10^3 = -0.0263 \text{ (USDm)}.$$

No terminal value. Staging USD3 m now and USD2 m in year 1, with cash in years 2–4, gives –USD0.2966 m.

Paying USD5 m now with a one-year cash delay gives –USD0.4784 m.

13 / Real and nominal consistency

Real and nominal three-year NPVs both –USD0.0263m

Illustrative example—not Olam data

$(1.10 \times 1.05) - 1 = 15.5\%$ nominal hurdle. Year 1: $2.10 / 1.155 = 2.00 / 1.10$.

Nominal cash grows by 5% inflation and is discounted at 15.5%. Project and currency risk require separate care.

Mixing nominal cash with a real hurdle overstates value.

14 / Operating versus investment break-even

Operating 33,333t; investment boundary 104,862 initial order tonnes

Illustrative example—not Olam data

$\text{USD2 m} / (\text{USD400} - \text{USD340 per tonne}) = 33,333 \text{ t}; / 150,000 \text{ t capacity} = 22.22\% \text{ utilization.}$

Full NPV includes ramp, tax, replacement, working cash and the finite horizon.

Operating above 22.22% utilization can still give negative investment NPV.

15 / Scenario weights are not likelihood

USD3m judgement-weighted mean

Illustrative example—not Olam data

Chosen NPVs [10, −8, 2] and weights [0.5, 0.3, 0.2]: $5 - 2.4 + 0.4 = \text{USD3 m.}$

The central USD10 m case is not a calibrated most-likely future. The adverse loss can breach funding.

If adverse weight is 0.5 and central weight 0.3, the mean becomes −USD0.6 m.

16 / Loss-tail VaR and CVaR

Loss VaR80 USD4m; tail CVaR80 USD8m

Illustrative example—not Olam data

NPVs $[-8, -4, 0, 2, 6]$ imply losses $[0, 0, 0, 4, 8]$. Nearest-rank VaR80 = 4; worst-20% CVaR80 = 8 (USDm).

Declare discrete tie treatment: $\text{mean}(\text{loss} \geq \text{VaR})$ would be USD6 m here, a different convention.

Stress code uses $\text{mean}(\text{loss} \geq 95\% \text{ quantile})$. Its chosen continuous generator has negligible ties.

17 / Common-exposure diversification

Expected total loss USD4m in both; joint tail differs

Illustrative example—not Olam data

Each source loses USD10 m with chosen probability 0.2. A shared port produces joint loss USD20 m with probability 0.2; independent joint probability is 0.04.

Chosen probabilities illustrate dependence. Country flags do not remove a shared crop season or port.

A different basin without a different port can leave route exposure unchanged.

18 / Difference in differences

20 illustrative incremental eligible sales units

Illustrative example—not Olam data

Treated sales: 100 → 130. Control: 100 → 110. DiD = $(130 - 100) - (110 - 100) = 20$.

Parallel trends and no material spillovers are required. Four chosen means cannot support an estimated interval.

At USD2 contribution/unit, gross incremental contribution is USD40 before service and pilot cost.

19 / Incremental ROMI

100% net ROMI; contribution/spend is 2.0 times

Illustrative example—not Olam data

Incremental contribution 120 – service 20 – trade 10 – cannibalisation 10 = 80. Marketing spend 40.
Net ROMI = $(80 - 40)/40 = 100\%$.

All inputs are chosen currency-thousands. Use a paid-sales counterfactual and full incremental costs.

If contribution after other costs is only 40, net ROMI is zero. Multi-year proposals need NPV.

20 / Decision value of information

USD2.5m net perfect-information ceiling

Illustrative example—not Olam data

Good state $p=0.5$: investment NPV 10; bad state $p=0.5$: -8 ; wait 0. Best prior $EV=1$; perfect-information $EV=5$; less study 1 and delay 0.5 gives net gain 2.5.

A perfect study is an upper bound. Do not add its benefit again to a policy that already includes learning.

An imperfect pilot may be worth less than its cost; keep the wait option.

21 / Reverse stress

About 4.86% more initial orders needed

Illustrative example—not Olam data

The managed owned template reaches zero NPV at 104,862 initial annual order tonnes. The current 100,000 t base needs 4.86% more initial orders.

The threshold depends on cost, hurdle, tax and capacity; volume cannot repair a negative unit contribution.

Cost, FX or commissioning shocks shift the boundary. Monitor orders together with cash.

22 / Size ranking can reverse

B has 3 times EBITDA despite half the category size

Illustrative example—not Olam data

Country A: demand 200 kt, capture 50 kt, unit contribution USD20, fixed USD0.7 m → EBITDA USD0.3 m. B: demand 100 kt, capture 40 kt, contribution USD40 → USD0.9 m.

A and B are fictional teaching cases. Chosen capex USD5 m vs USD2 m and NWC USD2 m vs USD0.5 m further favour testing B.

Before capital differences, A overtakes B above 80 kt captured at USD20/t while B stays at 40 kt.

Data construction and coverage

Missing observations remain unknown; identifiers and vintages are retained.

Join UN M49 identifiers to WDI by ISO3 and year; observers are VAT and PSE. Territories have a separate annex.

Convert WPP population from thousands to persons. A broad food-import proxy uses food share and merchandise imports from the same year.

Food production is a supply index; real PPP income, urban share and water stress are dated observations, not customer demand.

Complete-case regression uses no imputation. Macro availability does not establish candidate feasibility.

Demand and trade methods: no unsupported fitted elasticity

Implemented descriptive supply benchmark; product demand and trade models await data.

Demand begins with population and per-person quantity by segment; accessible customers, capture and delivery then bound sales.

AIDS/QUAIDS and discrete-choice models need observed spending, prices and identification. No imported elasticity is used.

A PPML gravity module would need bilateral product flows, fixed effects and barrier data. No fitted gravity result is claimed.

Aglink-Cosimo is an institutional benchmark. Its official projection is not extended mechanically to 2051.

Physical supply and adaptation need location-linked data

Do not turn a global warming number into a plant or crop loss coefficient.

A crop panel would link calendar-specific weather and basin conditions to yields, with fixed effects and validated nonlinear responses.

Convert fitted yield or downtime into saleable volume, procurement cost and stock days; price adaptation explicitly.

Traceability and standards affect access and compliance cost. Include a premium only when buyers pay it.

Asset coordinates, hazards and measured adaptation effects are unavailable; financial damage outputs remain conditional.

Operating and cash equations

Reconcile quantities and cash before computing returns.

Delivered volume = $\min(\text{eligible orders}, \text{usable capacity}, \text{route capacity})$. Revenue = delivered volume $\times$ net price.

EBITDA = volume $\times$ (price – variable cost per unit) – fixed cash cost. EBIT = EBITDA – D&A.

NWC = credit revenue $\times$ DSO/365 + inventory cost $\times$ DIO/365 – eligible purchases $\times$ DPO/365.

FCFF = EBIT – cash tax + D&A – capex – change in NWC + closeout. NPV discounts all incremental cash.

Mathematics disciplines the decision; it does not certify the future

Use the smallest method that answers the question and validates on relevant data.

Linear algebra supports regression and dependence; uncertainty requires an appropriate error structure.

Calculus and root finding locate margin and NPV thresholds. Capacity breakpoints can invalidate smooth derivatives.

Network flow connects origin, port and customer constraints. Inventory and queues need lead-time and service data.

Optimization, dynamic programming and Bayes structure actions and learning. Priors and transitions need evidence.

Executed OLS coefficients and uncertainty

No coefficient here is a country demand elasticity.

Term	Coefficient	Cluster SE	95% interval
Intercept	0.01852	0.00231	0.01398 to 0.02305
g_income_ppp	0.06197	0.03750	-0.01153 to 0.13546
du	0.19450	0.40929	-0.60770 to 0.99670
g_cereal_yield	-0.02331	0.01987	-0.06226 to 0.01563

SOURCE WB_food_index · WB_income_ppp · WB_urban · WB_cereal_yield

Holdout and transfer validation limit the model

Temporal improvement is small; geographic transfer is uneven.

Leave-region-out	Model MAE	Benchmark MAE
Asia	0.04888	0.04863
Africa	0.04444	0.04118
Europe	0.06458	0.06461
Americas	0.04561	0.04558
Oceania	0.03681	0.03757

Latest-vintage temporal validation is retrospective. The small error gain and uneven geographic transfer do not justify investment forecasting.

Stress specification and convergence

Simulation quantiles describe the chosen generator only.

Runs	p10 NPV	p50 NPV	Loss CVaR95
2500	-18.91	-1.65	29.14
5000	-19.16	-1.89	28.98
10000	-19.27	-1.65	29.25

Chosen latent factors jointly alter volume, costs, stocks and rates. Seed 20261008; covariance is constructed, not fitted to country data.

Prior art and originality boundary

Integration is the proposal; mathematical novelty is not asserted.

Robust regret, CVaR, constrained optimization, causal experiments, real options and EVSI are established methods.

Growth, climate, time-series and causal research guide module choice; no Nobel research parameter is imported into company forecasts.

The proposed integration links usable customer output to seasonal cash and transferable learning.

Only the transparent teaching comparison is executed. Superiority over real country allocation is unvalidated.

Pilot design and power: size from evidence

A convenient sample is not proof of an economically meaningful effect.

For two independent arms, $n \approx 2(z_{\alpha/2} + z_{\beta})^2 \sigma^2 / \delta^2$ per arm. Cluster design effect = $1 + (m - 1)ICC$.

Chosen $\sigma=10$, $\delta=5$, $\alpha=.05$ and 80% power imply about 63 observations per arm before clustering or attrition.

With 10 customers per cluster and $ICC=.10$, the design effect is 1.9: about 120 adjusted observations per arm.

Actual route variance, balance, spillovers and attrition determine the final sample. No actual sample size is approved.

Country atlas: all countries, six worlds, seven milestones

8,190 matrix rows preserve unknown finance rather than fabricate forecasts.

The country view links observations and source years to population paths, product/route hypotheses, modes and evidence gates.

Each of six worlds shows a normalized order threshold; country NPV remains unknown.

Seven milestones: 2026, 2030, 2035, 2040, 2045, 2050 and 2051. Exact matrix rows are downloadable.

Separate M49 territories are indexed. An undisclosed location is not proof that the company is absent.

Rank missing data by the decision it can reverse

The package is a complete public-data decision scaffold; internal investment cases remain open.

Orders, capture, prices and costs can reverse contribution, NPV and country priorities.

Monthly cash, FX, hedges and tax can create unfunded losses despite positive annual EBITDA.

Yield, capacity, route access and rights can remove a candidate from the feasible set.

Asset hazards, competitors and causal adoption can reverse resilience and transferability.

Refresh evidence when events or decisions change

Keep a dated reproducible version; do not silently revise the investment case.

Recheck company and ownership evidence at material transactions and results; preserve perimeter conflicts.

Legal rechecks rules, taxes, sanctions and rights before a named route commitment.

Review customer orders and stock cash monthly; escalate breaches immediately and test repeat after support ends.

Refresh data vintages, holdouts and scenarios annually; explain changed decisions, not only better fit.

Starting-deck audit: every original slide receives treatment

Detailed 44-row audit retains arithmetic and replaces unsupported certainty.

Preserve customer economics, cash discipline, the Nigeria laboratory and accurately bounded candidate experience.

Repair ownership, tonnage definitions, official projection horizons, procurement FX overlap and working-capital bases.

Expand to 195 countries, six worlds, full cash flow, causal pilot design and a linked technical package.

Replace unverified review counts, model probability claims and implied funding with evidence and actual validation.

Source register / 1–6

Exact URLs and limitations are in the linked source book.

ID	Source / pinpoint
CO_AR	Olam Agri Annual Report 2025 · Printed pp. 3, 10–11; CFO overview and performance charts
CO_FACT	Olam Agri factsheet, June 2026 · pp. 1–2; footprint, products, ownership
CO_OWN	Olam Group reorganisation / Olam Agri · Olam Agri section: 27 April 2026 sale and 30 June CFG share issuance
CO_ABOUT	Olam Agri About Us · At a glance, products and ownership blocks
OECD	OECD–FAO Agricultural Outlook 2026–2035 · Publication 29 June 2026; executive summary, methodology; DOI 10.1787/47874669-en
OECD_METHOD	Aglink-Cosimo methodology · Model overview and projection assumptions

Source register / 7-12

Exact URLs and limitations are in the linked source book.

ID	Source / pinpoint
UN_M49	UN Statistics Division M49 · Country/area codes and region/subregion table
UN_MEMBERS	UN Dag Hammarskjöld Library: current member states · All 193 country sections, crosswalked to UN M49 ISO3
UN_OBSERVERS	UN Dag Hammarskjöld Library: non-member observers · Holy See and State of Palestine sections
UN_WPP	UN World Population Prospects 2024 · Medium variant sheet; country rows at 2026, 2030, 2035, 2040, 2045, 2050, 2051
WB_population	World Bank WDI: SP.POP.TOTL · Source 2; series and country/year records, update 2026-07-13
WB_income_ppp	World Bank WDI: NY.GDP.PCAP.PP.KD · Source 2; series and country/year records, update 2026-07-13

Source register / 13–18

Exact URLs and limitations are in the linked source book.

ID	Source / pinpoint
WB_urban	World Bank WDI: SP.URB.TOTL.IN.ZS · Source 2; series and country/year records, update 2026-07-13
WB_inflation	World Bank WDI: FP.CPI.TOTL.ZG · Source 2; series and country/year records, update 2026-07-13
WB_food_index	World Bank WDI: AG.PRD.FOOD.XD · Source 2; series and country/year records, update 2026-07-13
WB_food_import_share	World Bank WDI: TM.VAL.FOOD.ZS.UN · Source 2; series and country/year records, update 2026-07-13
WB_food_export_share	World Bank WDI: TX.VAL.FOOD.ZS.UN · Source 2; series and country/year records, update 2026-07-13
WB_imports_usd	World Bank WDI: TM.VAL.MRCH.CD.WT · Source 2; series and country/year records, update 2026-07-13

SOURCE WB_urban · WB_inflation · WB_food_index · WB_food_import_share · WB_food_export_share · WB_imports_usd

Source register / 19–24

Exact URLs and limitations are in the linked source book.

ID	Source / pinpoint
WB_exports_usd	World Bank WDI: TX.VAL.MRCH.CD.WT · Source 2; series and country/year records, update 2026-07-13
WB_cereal_yield	World Bank WDI: AG.YLD.CREL.KG · Source 2; series and country/year records, update 2026-07-13
WB_water_stress	World Bank WDI: ER.H2O.FWST.ZS · Source 2; series and country/year records, update 2026-07-13
CO_NGA	Olam Agri Nigeria · Country capability/product sections
CO_GHA	Olam Agri Ghana · Country capability/product sections
CO_SEN	Olam Agri Senegal · Country capability/product sections

SOURCE WB_exports_usd · WB_cereal_yield · WB_water_stress · CO_NGA · CO_GHA · CO_SEN

Source register / 25–30

Exact URLs and limitations are in the linked source book.

ID	Source / pinpoint
CO_CMV	Olam Agri Cameroon · Country capability/product sections
CO_MOZ	Olam Agri Mozambique · Country capability/product sections
CO_ZAF	Olam Agri South-Africa · Country capability/product sections
CO_IND	Olam Agri India · Country capability/product sections
CO_VNM	Olam Agri Vietnam · Country capability/product sections
CO_BRA	Olam Agri Brazil · Country capability/product sections

Source register / 31–36

Exact URLs and limitations are in the linked source book.

ID	Source / pinpoint
IPCC	IPCC AR6 WGII Chapter 5: Food · Chapter 5; food system risks, adaptation and regional limits
WMO	WMO State of the Global Climate 2025 · 23 March 2026 release; global temperature and climate indicators
EUDR	EU Regulation 2023/1115 · Articles 3–4 and commodity/product annex
USDA_NGA	USDA Nigeria Grain and Feed Annual 2026 · Wheat supply/trade and flour-demand narrative
USDA_OIL	USDA Nigeria fiscal and trade policy changes 2026 · Oilseeds/products and crude/refined policy discussion
COMTRADE	UN Comtrade Nigeria wheat import preview · 2024, reporter 566, HS 1001, flow M, partner 0

Source register / 37-42

Exact URLs and limitations are in the linked source book.

ID	Source / pinpoint
CVAR	Rockafellar & Uryasev: Optimization of CVaR · Loss definition and CVaR optimisation formulation
DID	Callaway & Sant'Anna: Difference-in-differences · arXiv 1803.09015; identification with multiple time periods
NOBEL_2003	Economic Sciences prize 2003 · Prize motivation and research contribution
NOBEL_2018	Economic Sciences prize 2018 · Prize motivation and research contribution
NOBEL_2021	Economic Sciences prize 2021 · Prize motivation and research contribution
BASELINE	Portfolio baseline and résumé · <code>src/data/olamAfricaStrategy.ts</code> ; <code>src/lib/olamNigeriaDecisionModel.ts</code> ; <code>src/data/resume.ts</code>